

Jacob Sanders

Lead Software Engineer, Full-Stack & Applied AI

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SUMMARY

Lead software engineer and two-time founder with 13+ years shipping production systems end-to-end, from full-stack AWS serverless platforms to embedded firmware. Has directed and mentored engineering teams of 3–5 across multiple companies, and independently designed, built, and shipped SaaS platforms and AI tools on his own initiative. Builds production LLM applications on the Claude/Anthropic API, including multi-agent pipelines, prompt chaining, structured outputs, semantic search (RAG), and guardrails, with Claude Code as a daily development tool. Full-stack across Next.js/React/TypeScript and AWS serverless (Lambda, API Gateway, DynamoDB, Cognito, S3, IAM) provisioned as TypeScript infrastructure-as-code, with production relational-database experience (PostgreSQL via Supabase, T-SQL).

EXPERIENCE

Lien Into Revenue Services, LLC

Co-Founder

Huntington Beach, CA (Remote)

August 2025 – Present

- Co-founded and developed a production SaaS platform that manages the full lifecycle of personal injury chiropractic cases, from patient intake through lien negotiation and settlement collection
- Built the frontend in Next.js (App Router) and React, delivering a responsive role-based UI for chiropractors and case managers
- Designed the serverless AWS backend integrating DynamoDB, Lambda, AppSync (GraphQL), Cognito, S3, and CloudFront, defined as TypeScript infrastructure-as-code with AWS Amplify Gen 2 (CDK-based) and deployed through per-branch environments
- Implemented lien negotiation tooling enabling users to track outstanding liens, manage settlement offers, and monitor collection status across active cases

ARK Electronics

Full-Stack Engineer (1099 Contract)

Remote

February 2025 – April 2025

- Embedded with the client to scope requirements from scratch and ship a cloud-based manufacturing traceability platform end-to-end in weeks, covering product setup, serial number generation, barcode artifact creation, and test-result tracking for production and QA workflows
- Built a React/TypeScript/Vite (CoreUI) frontend and a fully serverless AWS backend using Lambda, AppSync (GraphQL), API Gateway (REST), DynamoDB, S3, and Cognito
- Exposed dual API surfaces (GraphQL for the web UI, REST for external clients) with API key management, usage plans, throttling, and event-driven job status updates for long-running generation batches

The Honeydew App, Inc.

Founder

Salt Lake City, UT (Remote)

April 2023 – February 2025

- Founded an on-demand handyman services marketplace; owned all system architecture and the majority of engineering while directing 4 University of Utah CS interns across frontend and backend workstreams
- Built a serverless TypeScript/Supabase (PostgreSQL) backend covering authentication, real-time job dispatching, and data persistence, paired with a Flutter (MVVM) consumer-facing mobile app
- Led UI/UX design in Figma in collaboration with an intern designer, producing complete user flows and high-fidelity mockups; delivered ~90% MVP completion before redirecting focus to a new venture

Nerd United

Blockchain Engineer

Lehi, UT (Hybrid)

April 2022 – April 2023

- Built and deployed a production generative-AI pipeline in Python, orchestrating the OpenAI image-generation API, IPFS pinning, and ERC-721 contract deployment as an automated multi-step workflow
- Developed blockchain applications in Go and Solidity on EVM-based protocols for cryptocurrency clients; deployed 8 GETH-based Proof-of-Authority testnets and 4 production mainnets, with security controls and access enforcement across infrastructure

Skullcandy

Firmware Architect

Park City, UT

October 2018 – March 2022

- Directed 3–5 firmware engineers through the full product lifecycle from prototype through mass-production release; served

as primary liaison to Apple’s MFi program office, translating technical constraints for non-technical stakeholders and cross-functional partners

- Architected firmware platform strategy across a portfolio of consumer audio products, establishing design patterns and standards that improved code quality and reduced per-product bring-up time
- Authored a C# desktop application for USB firmware updates and a Python CLI build-environment toolchain with automated regression tests, eliminating manual flashing and compressing nightly validation cycles

Code Corporation

Senior Firmware Engineer

Draper, UT

August 2013 – October 2018

- Led 3–4 firmware engineers building advanced barcode reading systems from board bring-up through production; drove Agile/Scrum adoption within the team
- Reduced device firmware upgrade time from ~3 min to ~60 sec via USB Mass Storage optimization, a 3× improvement in field servicing throughput
- Implemented a secure bootloader and BLE firmware for the Nordic nRF52832 SoC; contributed Linux kernel patches for image sensor, GPIO/PWM, and USB Mass Storage support; implemented Apple MFi chip authentication and the iAP2 protocol stack

TECHNICAL SKILLS

Languages	Python, TypeScript, Go, C, C++, C#, Swift, Kotlin
AI / LLM	Claude API (Anthropic SDK), Claude Code, multi-agent pipelines, prompt chaining, structured outputs, semantic search / RAG (Voyage AI embeddings), guardrails, LLM evals & backtesting, model tiering & cost optimization
Web / Frontend	Next.js, React, TypeScript, Flutter (MVVM), GraphQL, Figma
Cloud / AWS	Lambda, API Gateway, AppSync, DynamoDB, S3, CloudFront, Cognito, IAM roles & policies, AWS Amplify Gen 2 (TypeScript IaC on CDK)
Databases	PostgreSQL (Supabase), T-SQL / stored procedures, DynamoDB, relational & NoSQL data modeling
Dev Practices	Git, Docker, REST/GraphQL APIs, CI/CD, code review, automated testing (pytest, xUnit), Agile/Scrum team leadership

PROJECTS

Sentics: Multi-Agent Crypto Signal Pipeline

Python, Claude API, Next.js, Supabase, Vercel

Serverless multi-agent pipeline screening top crypto assets: a deterministic technical-analysis stage sets directional calls, a Claude synthesis stage writes rationale under a strict output whitelist (can’t override ground truth), and a second Claude stage distills news into structured sentiment reads. Model-tiered for cost (88% run-cost reduction) with a leak-free forward-tracking eval harness scoring live calls against realized returns (edge, hit-rate, Sharpe, t-stat).

github.com/jsanders5/sentics

Resumi: LLM Resume/Job-Matching Platform

Next.js, TypeScript, Claude API, Voyage embeddings

Full-stack app that scores job postings against a resume via the Claude API (structured JSON breakdowns across skills, experience, domain, requirements) and ranks matches with semantic search over Voyage embeddings. Prompt caching, backoff retries, and Sentry observability; deployed serverless on Vercel.

github.com/jsanders5/resumi

PySando: Interactive Python Learning Platform

FastAPI, React, Claude API, pytest

Full-stack tutorial that sandboxes and grades learner code against hidden pytest suites, paired with a Claude-powered tutor grounded in the live lesson, exercise, and learner’s code, guardrailed to never reveal solutions. Async multi-turn conversation state, provider abstraction, graceful degradation.

github.com/jsanders5/PySando

Shopify Inventory Sync & Storefront Alert (.NET / RAG)

ASP.NET Core 8, EF Core, T-SQL, Claude API, Voyage AI, xUnit

An ASP.NET Core 8 / EF Core Web API that HMAC-verifies real inventory webhooks and persists updates via a T-SQL stored procedure, paired with a storefront badge served through a reverse-proxied endpoint and verified end-to-end against a live store. Added a grounded product-Q&A feature (Voyage AI embeddings, Claude Haiku RAG) with a live grounded-vs-ungrounded comparison. Built with Claude Code in a structured spec/plan/review workflow; public repo, xUnit-tested. github.com/jsanders5/dotnet-shopify-demo

EDUCATION

University of Utah

M.S., Electrical & Computer Engineering • B.S., Computer Engineering

Salt Lake City, UT

2013